NARAYANA
IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | GTM-15 | Date: 09.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
D	A	D	D	C	B	D	D	B	C
11	12	13	14	15	16	17	18	19	20
B	A	D	B	D	C	D	D	D	D
21	22	23	24	25					
4	10	160	80	740					

CHEMISTRY

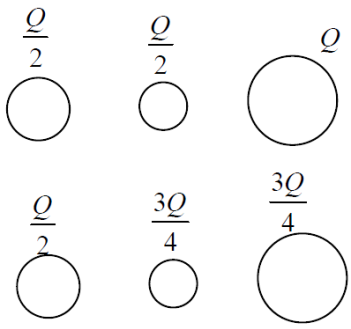
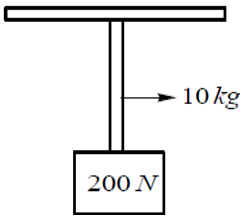
26	27	28	29	30	31	32	33	34	35
D	B	B	D	A	D	A	C	B	D
36	37	38	39	40	41	42	43	44	45
D	C	D	D	B	D	D	C	D	B
46	47	48	49	50					
543	12	0	287	17					

MATHEMATICS

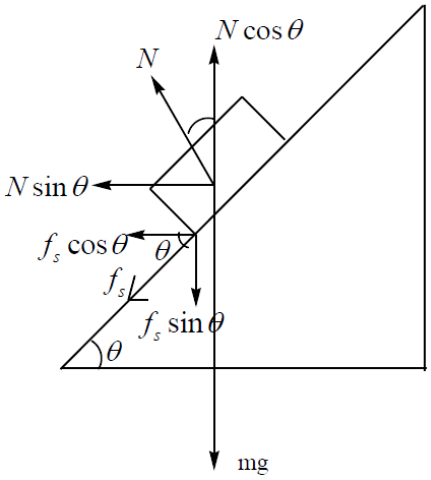
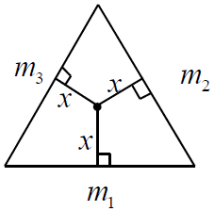
51	52	53	54	55	56	57	58	59	60
A	A	D	B	D	B	A	D	D	C
61	62	63	64	65	66	67	68	69	70
B	A	B	A	D	A	D	B	A	A
71	72	73	74	75					
43	61	71	38	17					

PHYSICS

1.	$-V = u - gt_1$ $-V = -u - gt_2$ $2u = g(t_1 - t_2)$ $u = \frac{g}{2}(t_1 - t_2) ----1$ $-H = ut_1 - \frac{1}{2}gt_1^2 ----2$ $-H = -ut_2 - \frac{1}{2}gt_2^2 ----3$ $-H = 0 - \frac{1}{2}gt^2 ----4$ $-gt^2 = u(t_1 - t_2) - \frac{1}{2}g(t_1^2 + t_2^2)$ On solving we get $t = \sqrt{t_1 t_2}$
2.	$\lambda = 3000\,A^\circ, \phi = 2.13\,eV$ $E = \frac{hc}{\lambda} = \frac{12400}{3000} = 4.13\,eV$ $E - \phi = 2\,eV \Rightarrow V_{SP} = 2V$
3.	$P_1 = \sqrt{2mk}$ $P_2 = \sqrt{2m36k} = 6P_1$ $\frac{\Delta P}{P} \times 100 = 500\%$
4.	For the equilibrium of any hemisphere of bubble $\Delta P \pi R^2 = S 4\pi R \Rightarrow \Delta P = \frac{4S}{R}$
5.	$x^2 = 1 + t^2 ---1$ Differentiate w.r.t 't' $2xv = 2t$ $xv = t ---2$ Again Differentiate w.r.t 't' $xa + v^2 = 1$ $a = \frac{1 - v^2}{x} = \frac{1 - \frac{t^2}{x^2}}{x} = \frac{x^2 - t^2}{x^3} = \frac{1}{x^3} = x^{-3}$ [$\because v = \frac{t}{x}$ From equation 2 and $x^2 - t^2 = 1$ from equation 1]
6.	$d = \frac{R}{4}$ $g' = g\left(1 - \frac{d}{R}\right) = g\left(1 - \frac{R}{4R}\right) = \frac{3}{4}g$ $mg' = mg\left(\frac{3}{4}\right) = 300 \times \frac{3}{4} = 225\,N$

7.	$S = \frac{Q}{m\Delta T}$ $S = L^2T^{-2}K^{-1}$ $gas\ cons\tan t = \frac{PV}{T} = \frac{ML^2T^{-2}}{K} = ML^2T^{-2}K^{-1}$
8.	$F = \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{d^2} = 16$  $F = \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{d^2} \times \frac{3}{8} = 16 \times \frac{3}{8} = 6$
9.	$P = VI$ $\frac{Q}{t} = I \Rightarrow I = \frac{ne}{t}$ $\frac{n}{t} = \frac{P}{Ve} = \frac{1}{2 \times 1.6} \times 10^9 = 31.25 \times 10^{17}$
10.	$f = 7$ $E = nf.\frac{1}{2}KT$ $E = \frac{7}{2}KT \times 10 = 35K_BT$
11.	$T = 200 + 10 \times 10 = 300\ N$ 
12.	$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$ $\lambda = \frac{1242}{6} = 207\ nm$
13.	$\mu = \frac{Real\ depth}{Apparant\ depth}$ $\mu = \frac{y_1 - y_3}{y_2 - y_3}$ From the data L. C = 0.001 cm $y_1 = 8.45 + 0.001 \times 26$ $y_2 = 7.12 + 0.001 \times 41$ $y_3 = 4.05 + 0.001 \times 1$

14.	$I=\frac{1}{2}\times E^2\times\varepsilon_0\times C=\frac{1}{2}\times 36\times 10^4\times 9\times 10^{-12}\times 3\times 10^8=486$
15.	$zero\,error=LC\times VC$ $0.04\,mm=LC\times 4$ $LC=0.01\,mm$ 49 MSD= 50 VSD 1 VSD = $\frac{49}{50}MSD$ $\therefore LC=1MSD-1VSD=\left(1-\frac{49}{50}\right)MSD=\frac{1}{50}MSD$ $0.01\,mm=\frac{1}{50}MSD$ 1 MSD = $\frac{1}{2}mm$ Number of MSD in 1 cm = $\frac{1cm}{\frac{1}{2}mm}=20$
16.	$\Delta Q=\Delta U+\Delta W$ $\Delta Q=nC_V\Delta T+\Delta W$ $48=1\times\frac{3R}{2}\times 2+\Delta W$ $48=3\times 8.3+\Delta W$ $\Delta W=48-24.9=23.1J$
17.	$\lvert e\rvert=L\frac{di}{dt}$ $0.1=L\times\frac{4}{0.2}$ $2\times 10^{-2}=4L$ $L=0.5\times 10^{-2}=5\times 10^{-3}H=5\,mH$
18.	$\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$ $\frac{1}{20}=(\mu-1)\left(\frac{1}{15}-\frac{1}{-30}\right)$ $\frac{1}{20}=(\mu-1)\frac{1}{10}$ $\mu=\frac{3}{2}=1.5$
19.	$\frac{1}{\lambda}=R\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$ $\frac{1}{\lambda}=1.097\times 10^7\left(\frac{1}{3^2}-\frac{1}{4^2}\right)$ $\frac{1}{\lambda}=1.097\times 10^7\times\frac{7}{9\times 16}$ $\lambda=\frac{9\times 16\times 10^{-7}}{1.097\times 7}=18.75\times 10^{-7}m=1.875\times 10^{-6}m$

20.	$N\sin\theta + f_s\cos\theta = \frac{mv^2}{r} ---1$ $N\cos\theta = mg + f_s\sin\theta ---2$ When V increases f_s also increases For no slipping $f_s \le \mu N$ $f_s = \mu N$ for maximum value of V Now equation 1 and 2 will become $N\sin\theta + \mu N\cos\theta = \frac{mv_{max}^2}{r} ---3$ $N\sin\theta + f_s\cos\theta = \frac{mv^2}{r} ---1$ $N\cos\theta = mg + f_s\sin\theta ---2$ When V increases f_s also increases For no slipping $f_s \le \mu N$ $f_s = \mu N$ for maximum value of V Now equation 1 and 2 will become $N\sin\theta + \mu N\cos\theta = \frac{mv_{max}^2}{r} ---3$ $if\ V > \sqrt{rg\tan\theta}$ 
21.	$x = \frac{h}{3} = \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}m$ $I = (m_1 + m_2 + m_3)x^2 = (2+4+6)\left(\frac{1}{\sqrt{3}}\right)^2 = 4kg - m^2$ 
22.	$f_0 = \frac{1}{2\pi\sqrt{LC}}$
23.	$i = \frac{24}{140.4 + \left(\frac{240\times10}{250}\right)} = 0.16A = 160mA$
24.	With air between plates $F = \frac{CV^2}{2d}$ With di-electrics between the plates and battery is remained across capacitor. $F^1 = \frac{K_e CV^2}{2d}, K_e = \frac{2+3}{2} = 2.5$ $8 = \frac{2.5\times10\times10^{-6}\times V^2}{2\times10\times10^{-3}}$ $V = 80volts$ Also if dielectrics are inserted after battery is removed. $F = \frac{CV^2}{2K_e d} \Rightarrow V = 200V$

25.	$t_1 - t_2 = \frac{6V}{2l_1} - \frac{5V}{2l_2} = \frac{V}{2}\left(\frac{6}{l_1} - \frac{5}{l_2}\right) = 740\,Hz$

CHEMISTRY

26.	Enzymes are biocatalyst and are highly specific. Oxidase enzyme oxidizes the groups but conversion of maltose to glucose is a hydrolysis reaction
27.	A) $N_2 + O_2 \rightarrow 2NO \Rightarrow$ it's a combination reaction B) $2Pb(NO_3)_2 \rightarrow 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$ - (I) This is a decomposition reaction. Reaction in which bigger molecules breaks down to simple unit is called decomposition reaction. C) $2Na_{(s)} + 2H_2O_{(\ell)} \rightarrow 2NaOH_{(aq)} + H_2$ - (II) This reaction is a displacement reaction. D) $2NO_2 + 2OH^{\Theta} \rightarrow NO_2^{\Theta} + NO_3^{\Theta} + H_2O$ In the above reaction we identify nitrogen undergoes both oxidation and reduction $\therefore$ This is a disproportionation reaction
28.	$\rightarrow$ As we down the group atomic radius will increase (A) $\rightarrow$ wrong $\rightarrow$ due to ineffective shielding of *d* & *f* electrons down the group E.N will remain almost same.
29.	Br^{Θ} ion will react with $AgNO_3$ and form pale yellow precipitate which is sparingly soluble in NH_4OH
30.	$PF_5 - sp^3d$ hybridization - trigonal bi pyramidal $BrF_5 - sp^3d^2$ hybridization square pyramidal Statement – I is wrong $SF_6 \rightarrow sp^3d^2$ hybridization-octahedral $\left[Co(NH_3)_6\right]^{+3} \rightarrow d^2sp^3$ hybridization. It is an inner orbital complex Statement –II is wrong
31.	$(A)Sc(21):[Ar]4s^2 3d^1 \rightarrow one\ unpaired\ e^{\Theta}$ $(B)Cr(24):[Ar]4s^1 3d^5 \rightarrow Six\ unpaired\ e^{\Theta}$ $(C)V(23):[Ar]4s^2 3d^3 \rightarrow Three\ unpaired\ e^{\Theta}$ $(D)Ti(22):[Ar]4s^2 3d^2 \rightarrow Two\ unpaired\ e^{\Theta}$ $(E)Mn(25):[Ar]4s^2 3d^5 \rightarrow five\ unpaired\ e^{\Theta}$ $\therefore A < D < C < E < B$

32.

$$CH_3-COOH \xrightarrow{(i)\,LiAlH_4} CH_3-CH_2-OH$$

$$\downarrow PCC$$

$$CH_3-CHO$$

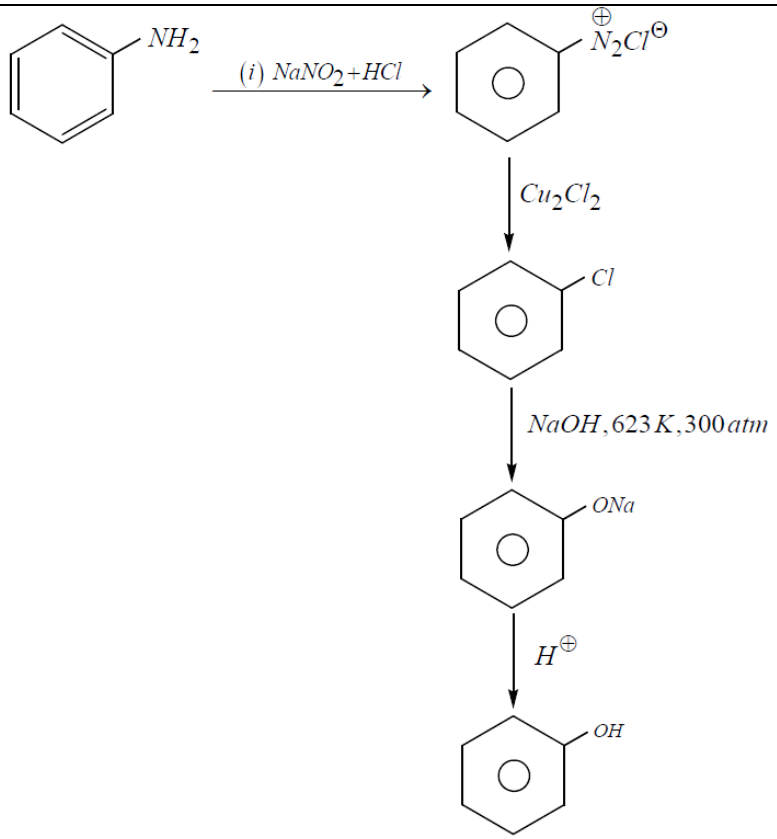
$$\downarrow HCN/OH^{\ominus}$$

$$CH_3-\underset{\underset{H}{|}}{\overset{\overset{OH}{|}}{C}}-CN$$

$$\downarrow H_2O/OH^{\ominus},\Delta$$

$$CH_3-\underset{\underset{}{|}}{\overset{\overset{OH}{|}}{CH}}-COOH$$

33.



34.

Given M = 3

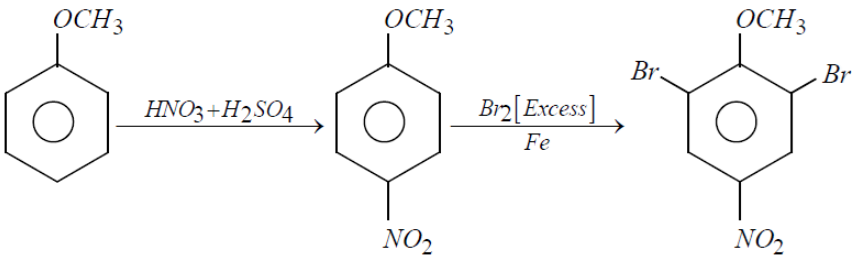
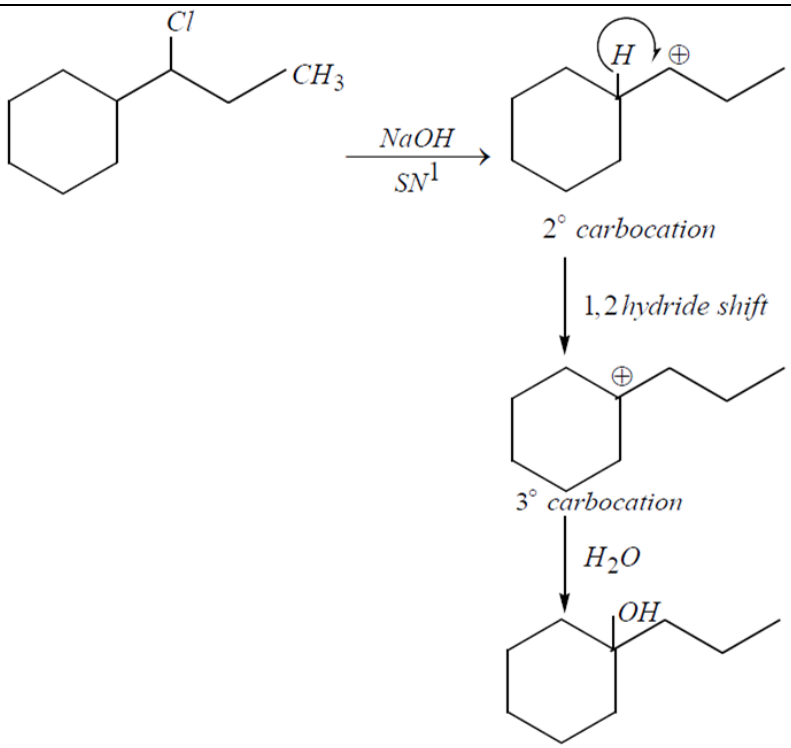
$d = 1.25\,g/mL$

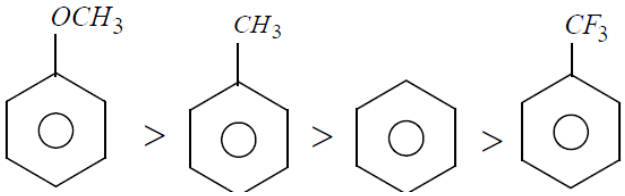
$m.wt\ of\ NaCl = 58.5$

$$m = \frac{M \times 1000}{d \times 1000 - M \times GMwt}$$

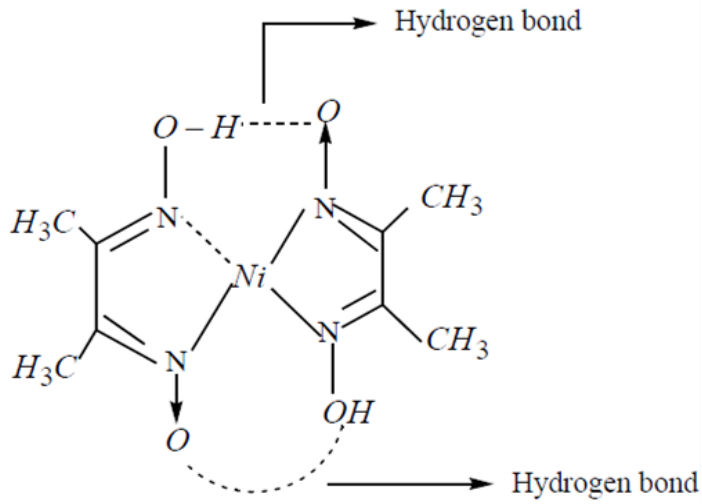
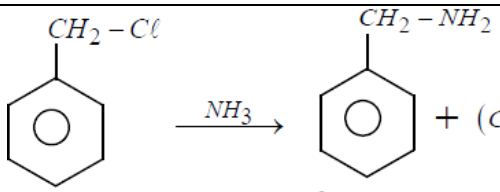
$$= \frac{3 \times 1000}{1.25 \times 1000 - 3 \times 58.5}$$

$$= \frac{3000}{1074.5} = 2.79\,m$$

35.	
36.	$CO_{(g)} + \frac{1}{2}O_{2(g)} \quad CO_{(g)}$ $K_P = K_C(RT)^{\Delta n}$ $\Delta n = n_P - n_R$ $= 1 - \left(1 + \frac{1}{2}\right)$ $= -\frac{1}{2}$ $K_P = K_C(RT)^{-1/2}$ $\frac{K_P}{K_C} = \frac{1}{\sqrt{RT}}$
37.	More polar compounds will have smaller R_f values since they will have a stronger affinity for the solid phase
38.	
39.	If a potential difference greater than the potential of the electrochemical cell is applied. In this case, the reaction begins to proceed in the opposite direction, that is, the nonspontaneous reaction takes place like in an electrolytic cell

40.	Given complex –Tetrahedral $TiCl_4; Ti^{+4} : d^0 : e^0 t^0$ $[FeO_4]^{-2}; e^2 t^0$ $[FeCl_4]^{\Theta}; Fe^{+3}; e^2 t^3$ $[CoCl_4]^{-2}; Co^{+2}; e^4 t^3$
41.	Pb^{+4} and $T\ell^{+3}$ act as oxidizing agents. The stability of higher oxidation state decreases down the group due to inert pair effect
42.	Reactivity towards aromatic electrophilic substitution reaction increases if electron releasing groups attached to aromatic ring reactivity: 
43.	Wave length of emission spectra (nm) $Li - 670.8,\ Na - 589.2,\ K - 766.5,\ Rb - 780.0,\ Cs - 455.5$
44.	$\left[PtBr_2\left(PMe_3\right)_2\right]$ Pt oxidation number = +2 Dibromobis(trimethylphosphine) platinum (II)
45.	Dipole moment of Cis-2-butene is greater than trans-2-butene $\mu_{(trans)} = 0$

46.	$\Delta T_f = 2.5$ $\Delta T_f = k_f \Rightarrow m = \frac{2.5}{1.88}$ $wt = \left(\frac{2.5}{1.86}\right) \times \frac{32}{10}$ $wt = 4.30\ gram$ $d = \frac{m}{V} \Rightarrow V = \frac{m}{d} = \frac{4.30}{0.792} = 5.43\ m\ell$ $\Rightarrow x \times 10^{-2} = 5.43 \Rightarrow x = 543$

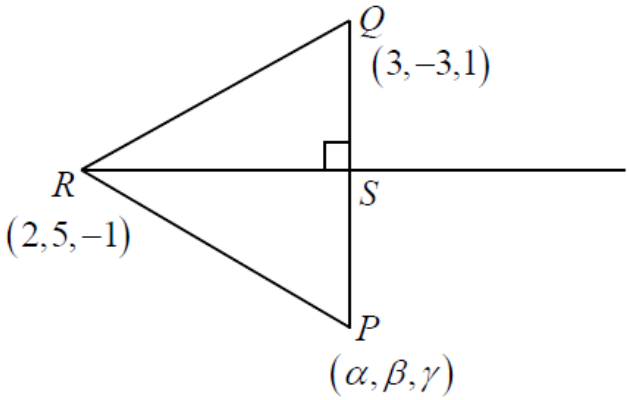
47.	
48.	**Oxidising power:** $MnO_4^- > Cr_2O_7^{-2} > VO_2^+$ $V^{+5}\left(d^0s^0\right); Mn^{+7}\left(d^0s^0\right); Cr^{+6}\left(d^0s^0\right)$
49.	 $+ \ (C_6H_5-CH_2)_2NH + (C_6H_5-CH_2)_3N$ $1^0-amine \quad 2^0-amine \quad 3^0-amine$ $3^0-amine$ + p-toluene sulphonyl chloride $\rightarrow$ No ppt (Clear solution) $\therefore$ M.wt of $3^0-amine \ (PhCH_2)_3N$ is 287
50.	$\frac{t_1}{t_2}=\frac{5}{2}\Rightarrow t_{1/2}=\frac{0.693}{t_1}$ $\Rightarrow t_{1/2}=\frac{0.693}{t_2}$ $t_1=\frac{2.303\times(t_{1/2})_1}{0.693}\log\frac{100}{(100-66.66)}=0.477$ $t_2=\frac{2.303\times(t_{1/2})_2}{0.693}\log\frac{100}{(100-8)}=0.699\text{k}$ $\frac{t_1}{t_2}=\frac{(t_{1/2})_1}{(t_{1/2})_2}.\frac{\log\left(\frac{100}{33.33}\right)}{\log\left(\frac{100}{20}\right)}=\frac{5}{2}\times\frac{\log 3}{\log 5}=\frac{0.477}{0.699}$ $\frac{5}{2}\times\frac{0.477}{0.699}\Rightarrow\frac{2.385}{1.398}=17$

MATHS

51. Let $\frac{x-0}{1}=\frac{y-3}{1}=\frac{z-1}{-1}=\lambda$

$x=\lambda, y=3+\lambda, z=1-\lambda$

$S=(\lambda,\lambda+3,-\lambda+1)$



d.r's of $SQ=(\lambda-3,\lambda+6,-\lambda)$

d.r's of line $=(1,1,-1)$

$\overline{SQ}\perp^{r}$ line $\Rightarrow 1(\lambda-3)+1(\lambda+6)-1(-\lambda)=0$

$\Rightarrow 3\lambda+3=0 \Rightarrow \lambda=-1$

$S=(-1,2,2)$

$(-1,2,2)=$ mid point of $PQ=\left(\frac{\alpha+3}{2},\frac{\beta-3}{2},\frac{\gamma+1}{2}\right)$

$-1=\frac{\alpha+3}{2}$ $\quad 2=\frac{\beta-3}{2}$ $\quad 2=\frac{\gamma+1}{2}$

$\alpha=-5$ $\quad \beta=7$ $\quad \gamma=3$

$\therefore P=(-5,7,3)$

Area of $\Delta PQR=\frac{1}{2}\left|\overline{PR}\times\overline{PQ}\right|$

$\overline{PQ}=(+8,-10,-2)$

$\overline{PR}=(7,-2,-4)$

	area of $\Delta PQR = \frac{1}{2}\left\|\overline{PR} \times \overline{PQ}\right\|$ $\lambda = \frac{1}{2}\begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 8 & -10 & -2 \\ 7 & -2 & -4 \end{vmatrix}$ $= \frac{1}{2}\left\|\bar{i}(40-4) - \bar{j}(-32+14) + \bar{k}(-16+70)\right\|$ $= \frac{1}{2}\left\|36\bar{i} + 18\bar{j} + 54\bar{k}\right\|$ $\lambda = \frac{1}{2}\sqrt{1296+324+2916} = \frac{1}{2}\sqrt{4536}$ $\lambda^2 = \frac{4536}{4} = 1134$ $14K = \frac{4536}{4} = 1134$ $K = \frac{4536}{4} = 81$
52.	$m, m-4, m-8, ----, m-96$ for 25 days $sum = m + (m-4) + (m-8) + ------ + (m-96)$ $17m = \frac{25}{2}\left[2m + 24(-4)\right]$ $34m = 50m - 25 \times 96$ $25 \times 96 = 16m \quad \Rightarrow m = 150$
53.	$\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ $\Rightarrow \beta x\, dy - 2\alpha y\, dy - (\beta\gamma - 4\alpha)dy = (2+\alpha)x\,dx - \beta y\,dx + 2\,dx$ $\Rightarrow (2+2)x\,dx - \beta(x\,dy + y\,dx) + 2\,dx + 2\alpha y\,dy + (\beta\gamma - 4\alpha)dy = 0$ Integrating on both sides $\Rightarrow (2+\alpha)\int x\,dx - \beta\int d(xy) + 2\int 1dx + 2\alpha\int y\,dy + (\beta\gamma - 4\alpha)\int 1dy = 0$ $\Rightarrow (2+\alpha)\frac{x^2}{2} - \beta xy + 2x + 2\alpha\frac{y^2}{2} + (\beta\gamma - 4\alpha)y = c ----(1)$

	This equation represent a circle coefficient of x^2 = coefficient of y^2 $\Rightarrow \frac{2+\alpha}{2}=\alpha \Rightarrow 2+\alpha=2\alpha$ $\alpha=2$ coefficient of $xy=0 \Rightarrow \beta=0$ $(1) \Rightarrow 2x^2+2y^2+2x-8y=c$ since circle passes $(0,0)$ $c=0$ $\therefore$ circle $2x^2+2y^2+2x-8y=0$ $\Rightarrow x^2+y^2+x-4y=0$ $c=\left(-\frac{1}{2},2\right)$ $r=\sqrt{\frac{1}{4}+4}=\sqrt{\frac{17}{4}}=\frac{\sqrt{17}}{2}$
54.	α,β be roots of $x^2+\sqrt{2}x-8=0$ $U_n=\alpha^n+\beta^n$, then $\frac{U_{10}+\sqrt{2}U_9}{2U_8}$ α is a root of $x^2+\sqrt{2}x=8$ $\alpha^2+\sqrt{2}\alpha=8$ $\alpha^8\left(\alpha^2+\sqrt{2}\alpha\right)=\alpha^8 8$ $\alpha^{10}+\sqrt{2}\alpha^9=8\alpha^8$ Similarly $\beta^{10}+\sqrt{2}\beta^9=8\beta^8$ Add $\alpha^{10}+\beta^{10}+\sqrt{2}\left(\alpha^9+\beta^9\right)=8\left(\alpha^8+\beta^8\right)$ $U_{10}+\sqrt{2}U_9=8U_8$ $\frac{U_{10}+\sqrt{2}U_9}{2U_8}=\frac{8}{2}=4$
55.	$=\underset{n\to\infty}{Lt}\frac{n\left[\left(1^2-1\right)+\left(2^2-2\right)+.....+\left((n-1)^2-(n-1)\right)\right]-\left(-1\left(1^2-1\right)-2\left(2^2-2\right).....-(n-1)\left((n-1)^2-(n-1)\right)\right)}{\left(\sum n^3\right)-\left(\sum n^2\right)}$ $=\underset{n\to\infty}{Lt}\frac{n\left(\frac{(n-1)n(2n-1)}{6}-\frac{n(n-1)}{2}\right)-\left(\left(\frac{(n-1)n}{2}\right)^2-\frac{n(n+1)(2n+1)}{6}\right)}{\frac{n^2(n+1)^2}{4}-\frac{n(n+1)(2n+1)}{6}}$ $=\frac{\left(\frac{2}{6}\right)-\left(\frac{1}{2}\right)^2}{1/4}=\frac{\frac{1}{3}-\frac{1}{4}}{\frac{1}{4}}=\frac{\left(\frac{1}{12}\right)}{\left(\frac{1}{4}\right)}=\frac{1}{3}$

56.	$\lvert\bar{a}\rvert = \sqrt{36+1+1}$ $= \sqrt{38}$ $\lvert\bar{b}\rvert = \sqrt{1+1} = \sqrt{2}$ $\lvert\bar{c}-\bar{a}\rvert = 2\sqrt{2}$ S.O.B.S $\lvert\bar{c}\rvert^2 + \lvert\bar{a}\rvert^2 - 2\bar{c}\cdot\bar{a} = 8$ $\lvert\bar{c}\rvert^2 + 38 - 12\lvert\bar{c}\rvert = 8$ $\lvert\bar{c}\rvert^2 - 12\lvert\bar{c}\rvert + 30 = 0$ $\lvert\bar{c}\rvert = \frac{12 \pm \sqrt{144-120}}{2}$ $= \frac{12 \pm 2\sqrt{6}}{2}$ $\lvert\bar{c}\rvert = 6 \pm \sqrt{6}$ $\therefore\ \lvert\bar{c}\rvert = 6 + \sqrt{6}$ $\cos(\bar{a},\bar{b}) = \frac{\bar{a}\cdot\bar{b}}{\lvert\bar{a}\rvert\lvert\bar{b}\rvert} = \frac{6+1}{\sqrt{38}\sqrt{2}}$ $= \frac{7}{\sqrt{76}}$ $\sin(\bar{a},\bar{b}) = \sqrt{1-\cos^2(\bar{a},\bar{b})}$ $= \sqrt{1-\frac{49}{76}}$ $= \sqrt{\frac{27}{76}}$

	$=\frac{3\sqrt{3}}{2\sqrt{19}}$ $\bar{a}=6\hat{i}+\hat{j}-\hat{k}$, $\bar{b}=\hat{i}+\hat{j}$ $\bar{a}\cdot\bar{c}=6\lvert\bar{c}\rvert$ $\lvert\bar{c}-\bar{a}\rvert=2\sqrt{2}$, $(\bar{a}\times\bar{b},\bar{c})=60^0$ $\lvert(\bar{a}\times\bar{b})\times\bar{c}\rvert=\lvert\bar{a}\times\bar{b}\rvert\lvert\bar{c}\rvert\sin(\bar{a}\times\bar{b},\bar{c})$ $=\lvert\bar{a}\rvert\lvert\bar{b}\rvert\sin(\bar{a},\bar{b})\ \lvert\bar{c}\rvert\sin 60^0$ $=\sqrt{38}\ \sqrt{2}\ \frac{\sqrt{27}}{\sqrt{76}}(6+\sqrt{6})\frac{\sqrt{3}}{2}$ $=\sqrt{19}\sqrt{2}\sqrt{2}\frac{3\sqrt{3}}{2\sqrt{19}}\times(6+\sqrt{6})\frac{\sqrt{3}}{2}$ $=3\times3\times\frac{(6+\sqrt{6})}{2}$ $=\frac{9}{2}(6+\sqrt{6})$
57.	$\lvert z_1-2z_2\rvert=\lvert 1-2\bar{z}_1z_2\rvert$ S.O.B.S $\lvert z_1-2z_2\rvert^2=\lvert 1-2\bar{z}_1z_2\rvert^2$ $\boxed{\lvert z\rvert^2=z\cdot\bar{z}}$ $(z_1-2z_2)(\bar{z}_1-2\bar{z}_2)=(1-2\bar{z}_1z_2)(1-2z_1\bar{z}_2)$ $\Rightarrow\lvert z_1\rvert^2+4\lvert z_2\rvert^2-2z_1\bar{z}_2-2\bar{z}_1z_2$ $=1-2z_1\bar{z}_2-2\bar{z}_1z_2+4\lvert z_1\rvert^2\lvert z_2\rvert^2$ $\Rightarrow\lvert z_1\rvert^2+4\lvert z_2\rvert^2-4\lvert z_1\rvert^2\lvert z_2\rvert^2-1=0$ $\Rightarrow(\lvert z_1\rvert^2-1)(4\lvert z_2\rvert^2-1)=0$ $\Rightarrow\lvert z_1\rvert=1,\lvert z_2\rvert=\frac{1}{2}$
58.	$\underline{\lvert 5}\quad\underline{\lvert 4}\quad\underline{\lvert 3}\quad\underline{\lvert 2}\quad\underline{\lvert 1}\quad\underline{\lvert 0}$ $X\quad X\quad X\quad X\quad X\quad X$ $2\quad 0\quad 0\quad 0\quad 0\quad 1$ $\underline{\lvert 5}\times2+\underline{\lvert 0}\times1=240+1=241$ $315-241=74$ $74=\underline{\lvert 4}\times3+\underline{\lvert 2}\times1$ $\therefore$ NRAPGU is correct answer

59.

$$f(x)=\frac{1}{7-\sin 5x}$$

$$-1\le -\sin 5x\le 1$$

$$7-1\le 7-\sin 5x\le 7+1$$

$$6\le 7-\sin 5x\le 8\Rightarrow \frac{1}{8}\le \frac{1}{7-\sin 5x}\le \frac{1}{6}$$

$$Range=\left[\frac{1}{8},\frac{1}{6}\right]$$

60.

$$f(x)=\left(\frac{1}{x}\right)^{2x};x>0$$

Given : f attains the maximum at $x=\frac{1}{e}$

$\therefore$ f is increasing in $\left(\frac{1}{\pi},\frac{1}{e}\right)$

$$\therefore f\left(\frac{1}{\pi}\right)<f\left(\frac{1}{e}\right)$$

$$\left(\frac{1}{\frac{1}{\pi}}\right)^{2\times\frac{1}{\pi}}<\left(\frac{1}{\frac{1}{e}}\right)^{2\times\frac{1}{e}}\Rightarrow \pi^{\frac{2}{\pi}}<e^{\frac{2}{e}}$$

$$\Rightarrow\left(\pi^{\frac{2}{\pi}}\right)^{\frac{\pi e}{2}}<\left(e^{\frac{2}{e}}\right)^{\frac{\pi e}{2}}$$

$$\Rightarrow \pi^{e}<e^{\pi}$$

61.

	1	2	3	4	5
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)

m = upper triangle elements including diagonal elements + (5,4) = 16

n = Lower triangle elements excluding (5, 4) = 9

$\therefore m + n = 16 + 9 = 25$

62.	$\int_0^{\sqrt{1}}\left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx+\int_{\sqrt{1}}^{\sqrt{2}}\left[\left[x^2\right]+\left[\frac{x^2}{2}\right]\right]dx+\int_{\sqrt{2}}^{\sqrt{3}}\left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx$ $+\int_{\sqrt{3}}^{\sqrt{4}}\left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx++\int_{\sqrt{4}}^{\sqrt{5}}\left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx++\int_{\sqrt{5}}^{\sqrt{6}}\left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx$ $+\int_{\sqrt{6}}^{\sqrt{7}}\left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx++\int_{\sqrt{7}}^{\sqrt{8}}\left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx++\int_{\sqrt{8}}^{\sqrt{9}}\left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx$ $=\int_0^1 0dx+\int_1^{\sqrt{2}}1dx+\int_{\sqrt{2}}^{\sqrt{3}}3dx+\int_{\sqrt{3}}^{\sqrt{4}}4dx+\int_{\sqrt{4}}^{\sqrt{5}}6dx+\int_{\sqrt{5}}^{\sqrt{6}}7dx+\int_{\sqrt{6}}^{\sqrt{7}}9dx++\int_{\sqrt{7}}^{\sqrt{8}}10dx++\int_{\sqrt{8}}^{\sqrt{9}}12dx$ $=31-6\sqrt{2}-\sqrt{3}-\sqrt{5}-\sqrt{7}-2\sqrt{6}$ $a=31,b=-6,c=-2+$ $a+b+c=31-6-2=23$
63.	$\left\|\left(adj\left(-4adj\left(-3adj\left(3adj(2A)^{-1}\right)\right)\right)\right)\right\|^2$ $\left\|-4adj\left(-3adj\left(3adj(2A)^{-1}\right)\right)\right\|^2$ $4^6\left\|adj\left(-3adj\left(3adj(2A)^{-1}\right)\right)\right\|^2$ $4^6\left(\left\|\left(-3adj\left(3adj(2A)^{-1}\right)\right)\right\|^2\right)^2$ $4^6\left\|-3adj\left(3adj(2A)^{-1}\right)\right\|^4$ $4^6\times3^{12}\left\|adj\left(3adj(2A)^{-1}\right)\right\|^4$ $4^6\times3^{12}\left\|\left(3adj(2A)^{-1}\right)\right\|^8$ $4^6\times3^{12}\times3^{24}\left\|adj(2A)^{-1}\right\|^8$ $4^6\times3^{36}\left\|(2A)^{-1}\right\|^{16}$ $4^6\times3^{36}\times\frac{1}{2^{48}\|A\|^{16}}$

	$2^{12}\times 3^{36}\times\frac{1}{2^{48}\times 3^{16}}$ $2^{-36}\times 3^{20}=2^{m}\times 3^{n}$ $m=-36,\ n=20$ $m+2n=-36+2(20)$ $=-36+40$ $=4$
64.	${}^{(n+1)}C_{r+1}:{}^{n}C_{r}:{}^{(n-1)}C_{r-1}=55:35:21$ $\frac{n+1}{r+1}\ {}^{n}C_{r}:{}^{n}C_{r}:\frac{r}{n}\,{}^{n}C_{r}$ $\frac{n+1}{r+1}:1:\frac{r}{n}=55:35:21$ $\frac{n+1}{r+1}=\frac{55}{35}=\frac{11}{7}$ $7n-11r=14$ .....(1) $\frac{n}{r}=\frac{35}{21}=\frac{5}{3}$ $3n-5r=0$.....(2) From (1) and (2) $n=10, r=6$ $2n+5r=2\times 10+5\times 6=20+30=50$
65.	$\int\frac{1}{a^2\sin^2 x+b^2\cos^2 x}dx=\frac{1}{12}\tan^{-1}(3\tan x)+c$ $L.H.S=\int\frac{1}{a^2\sin^2 x+b^2\cos^2 x}dx=\int\frac{\sec^2 x}{a^2\tan^2 x+b^2}dx=\frac{1}{a^2}\int\frac{\sec^2 x}{\tan^2 x+\left(\frac{b}{a}\right)^2}dx$ Let $\tan x=t\Rightarrow\sec^2 x dx=dt$ $=\frac{1}{a^2}\int\frac{1}{t^2+\left(\frac{b}{a}\right)^2}dx$ $=\frac{1}{a^2}.\frac{1}{\frac{b}{a}}\tan^{-1}\left(\frac{t}{\frac{b}{a}}\right)+c$ $LHS=\frac{1}{ab}.\tan\left(\frac{a}{b}\right)+c=\frac{1}{ab}\tan^{-1}\left(\frac{a\tan x}{b}\right)+c$ $RHS=\frac{1}{12}.\tan^{-1}(3\tan x)+c$ $ab=12\Rightarrow\frac{a}{b}=3$ $a^2=36\rightarrow b^2=4$ $\therefore$ Max value of $a\sin x+b\cos x=\sqrt{a^2+b^2}=\sqrt{36+4}=\sqrt{40}$

66.	Given points are (2,1) and (1,3) Take point $P(x, y)$ $\overset{5}{\vdash\!\!\!-\!\!\!-\!\!\!-

68.

$$A=\int_1^2\left(\frac{1}{x}-\frac{a}{x^2}\right)dx=\left(\log|x|+\frac{a}{x}\right)_1^2$$

$$A=\log 2-\frac{1}{7}$$

$$\log 2-\frac{1}{7}=\log 2+\frac{a}{2}-\log 1-a$$

$$\Rightarrow\ \log 2-\frac{1}{7}=\log 2-\frac{a}{2}$$

$\therefore$ equating on both sides

$$\Rightarrow\frac{1}{7}=\frac{a}{2}\Rightarrow a=\frac{2}{7}$$

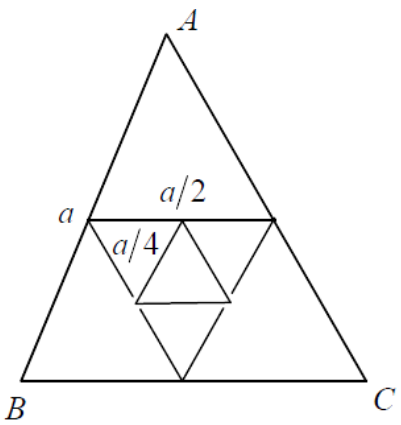
Consider $7a-3=7\left(\frac{2}{7}\right)-3$

$=2-3=-1$

69.

$$P=3a+3\left(\frac{a}{2}\right)+3\left(\frac{a}{4}\right)+-----$$

$$Q=\frac{\sqrt{3}}{4}a^2+\frac{\sqrt{3}}{4}\left(\frac{a}{2}\right)^2+\frac{\sqrt{3}}{4}\left(\frac{a}{4}\right)^2+-----$$



$$P=3a\left(1+\frac{1}{2}+\frac{1}{4}+----\right)$$

$$P=3a\left(\frac{1}{1-\frac{1}{2}}\right)$$

$$P=3a(2)=6a$$

$$P^2=36a^2$$

$$Q=\frac{\sqrt{3}}{4}a^2\left(1+\frac{1}{4}+\frac{1}{16}+-----\right)$$

$$Q=\frac{\sqrt{3}}{4}a^2\left(\frac{1}{\frac{3}{4}}\right)$$

$$Q=\frac{\sqrt{3}}{4}a^2\cdot\frac{4}{3}$$

$$Q=\frac{a^2}{\sqrt{3}}$$

$$36\sqrt{3}Q=36\sqrt{3}\cdot\frac{a^2}{\sqrt{3}}$$

$$=36a^2$$

Hence $P^2=36\sqrt{3}Q$

70. $\therefore$ 3 letters can be posted to any one of the 5 address

$$= 5^3$$

$\therefore$ 3 letters can be posted to exactly two address is

$$^5C_2\left(2^3-2\right)$$

Required probability

$$=\frac{^5C_2(8-2)}{5\times5\times5}$$

$$=\frac{5\times2\times6}{5\times5\times5}=\frac{12}{25}$$

71. $L_1:\frac{x-\lambda}{3}=\frac{y-2}{-1}=\frac{z-1}{1}$ and $L_2:\frac{x+2}{-3}=\frac{y+5}{2}=\frac{z-4}{4}$

$$S.D=\frac{44}{\sqrt{30}}$$

$$A(\lambda,2,1),B(-2-5,4)$$

$$P=3i-j+k \qquad q=-3i+2j+4k$$

$$p\times q=\begin{vmatrix} i & j & k \\ 3 & -1 & 1 \\ -3 & 2 & 4 \end{vmatrix}=i(-4-2)-j(12+3)+k(6-3)$$

$$p\times q=-6i-15j+3k$$

$$|p\times q|=\sqrt{36+225+9}=\sqrt{270}=3\sqrt{30}$$

$$AB=(\lambda+2)i+7j-3k$$

$$S.D=\frac{|AB.(p\times q)|}{|p\times q|}=\frac{|1(\lambda+2)(-6)-105-9|}{3\sqrt{30}}$$

$$\frac{44}{\sqrt{30}}=\frac{|-6\lambda-12-105-9|}{3\sqrt{30}}$$

$$3\times44=1-6\lambda-1261\Rightarrow1-6\lambda-1261=132$$

$$-6\lambda-126=132,-6\lambda-126=-132$$

$$-6\lambda=258,-6\lambda=-6$$

$$\lambda=-43 \qquad \lambda=1$$

Largest value of $|\lambda|=43$

72. **Note:** Insufficient data, given line is not a tangent to the hyperbola thus formed with the given data

73.

X	0	1	2	3
$P(X=xi)$	$\frac{^9C_5}{^{12}C_5}$	$\frac{^9C_4\times{}^3C_1}{^{12}C_5}$	$\frac{^9C_3\times{}^3C_2}{^{12}C_5}$	$\frac{^9C_2\times{}^3C_3}{^{12}C_5}$

$$Mean=\sum x_iP(X=x_i)$$

	$=0+1\left(\frac{{}^{9}C_4 \times {}^{3}C_1}{{}^{12}C_5}\right)+2\left(\frac{{}^{9}C_3 \times {}^{3}C_2}{{}^{12}C_5}\right)+31\left(\frac{{}^{9}C_2 \times {}^{3}C_3}{{}^{12}C_5}\right)$ $=0+\frac{21}{44}+\frac{2(14)}{44}+\frac{2(3)}{44}=\frac{55}{44}=\frac{5}{4}$ $\text{variance}=\sum xi^2 P(x=xi)-\mu^2$ $=0+\frac{21}{44}+\frac{4\times 14}{44}+9\times\frac{2}{44}-\left(\frac{5}{4}\right)^2$ $=\frac{21}{44}+\frac{56}{44}+\frac{18}{44}-\frac{25}{16}=\frac{95}{44}-\frac{25}{16}$ $=\frac{1520-1100}{44\times 16}=\frac{420}{16\times 44}=\frac{105}{176}=\frac{m}{n}$ $n-m=176-105=71$
74.	$2x+7y+\lambda z=3$ $3x+2y+5z=4$ $x+\mu y+32z=-1$ $\Delta=0$ $\begin{vmatrix} 2 & 7 & -1 \\ 3 & 2 & 5 \\ 1 & \mu & 32 \end{vmatrix}=0$ $2(64-5\mu)-7(96-5)-1(3\mu-2)=0$ $128-10\mu-637-3\mu+2=0$ $-507=13\mu \Rightarrow \mu=-39$ $\Delta_2=\begin{vmatrix} 2 & 3 & \lambda \\ 3 & 4 & 5 \\ 1 & -1 & 32 \end{vmatrix}=0$ $2(128+5)-3(96-5)+\lambda[-3-4]=0$ $266-273-7\lambda=0$ $-7-7\lambda=0$ $-7\lambda=7 \Rightarrow \lambda=-1$ $\lambda-\mu=-1-(-39)=38$

75.	$f(x)=\left[\frac{x}{2}\right]-\left[\sqrt{x}\right]+3$ $[0,8]$ $x=1,2,4,6,8$ at $\boxed{x=1}$ $L.H.L=0-0+3=3$ $R.H.L=0-1+3=2$ $L.H.L\neq R.H.L$ discontinuous at x = 1 $\boxed{x=2}$ $L.H.L=0-1+3=2$ $R.H.L=1-1+3=3$ $L.H.L\neq R.H.L$ discontinuous at x = 2 $\boxed{x=4}$ $L.H.L=1-1+3=3$ $R.H.L=2-2+3=3$ $L.H.L=R.H.L=f(4)$ continuous at x = 4 $\boxed{x=6}$ $L.H.L=2-2+3=3$ $R.H.L=3-2+3=4$ $L.H.L\neq R.H.L$ discontinuous at x = 6 $\boxed{x=8}$ $L.H.L=3-2+3=4$ $f(8)=4-2+3=5$ $L.H.L\neq R.H.L$ discontinuous at x = 8 $\therefore$ points of discontinuity x = 1, 2, 6, 8 sum = 1 + 2 + 6 + 8 =17